QUESTION ANALYSIS

EEE455

VLSI Design and Testing

Question Analysis

Sessions 2020–21 to 2019–20 *2 papers, 53 questions, 90 formulas*

INSIDE

• At a glance • Heat map • Priority list • 12 topic pages • Derivations • Repeats • Styles • Study order

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1. At a glance:

This analysis covers **2 papers** (2020-21, 2019-20), with 53 questions and 294 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Combinational logic styles	49	17 %	10	Stick diagrams (12); Static gates (9); Dynamic logic (9)
Delay and logical effort	46	16 %	8	Path delay (25); Gate delay (7); Input ordering (5)
Sequential circuits and timing	36	12 %	6	Setup and hold (10); Flip-flops (10); Hold time (6)
Fabrication	30	10 %	6	Lithography (16); Etching (6); Cleaning (6)
Power	28	10 %	5	Dynamic power (16); Glitches (6); Leakage (6)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2020-21	2019-20	All
MOS transistor theory	**4**	**13**	**17**
I-V equations	·	**8**	8
Leakage and DIBL	·	**5**	5
C-V characteristics	**4**	·	4
CMOS inverter	**13**	**7**	**20**
DC characteristics	·	**7**	7
Skew	**7**	·	7
Beta ratio	**6**	·	6
Fabrication	**10**	**20**	**30**
Lithography	**8**	**8**	16
Etching	·	**6**	6
Cleaning	·	**6**	6
Diffusion and implantation	**2**	·	2
Delay and logical effort	**21**	**25**	**46**
Path delay	**12**	**13**	25
Gate delay	·	**7**	7
Input ordering	·	**5**	5
Elmore delay	**5**	·	5
Sizing	**4**	·	4
Power	**12**	**16**	**28**
Dynamic power	**12**	**4**	16
Glitches	·	**6**	6
Leakage	·	**6**	6
Interconnect	·	**10**	**10**
Crosstalk	·	**7**	7

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Topic and subtopic	2020-21	2019-20	All
Sheet resistance	·	**3**	3
Scaling	·	**11**	**11**
Moore's law	·	**6**	6
Scaling rules	·	**5**	5
Combinational logic styles	**24**	**25**	**49**
Stick diagrams	**6**	**6**	12
Static gates	·	**9**	9
Dynamic logic	**3**	**6**	9
Pseudo-nMOS	**8**	·	8
Pass transistors	**7**	·	7
CVSL	·	**4**	4
Sequential circuits and timing	**26**	**10**	**36**
Setup and hold	**10**	·	10
Flip-flops	**10**	·	10
Hold time	·	**6**	6
FSM	**6**	·	6
Cycle time	·	**4**	4
Datapath	**7**	·	**7**
Bus architectures	**7**	·	7
Memories and arrays	**13**	**13**	**26**
SRAM	**10**	**6**	16
ROM	·	**7**	7
PLA	**3**	·	3
Verilog and design flow	**14**	·	**14**
Verilog	**8**	·	8
Design flow	**6**	·	6

3. Priority list:

Must know: Combinational logic styles (49), Delay and logical effort (46), Sequential circuits and timing (36), Fabrication (30), Power (28), Memories and arrays (26), CMOS inverter (20), MOS transistor theory (17)

Likely: Verilog and design flow (14), Scaling (11), Interconnect (10), Datapath (7)

Less often: none

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

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4. Topic pages:

4.1. MOS transistor theory:

17 marks, 3 questions, in 2 of 2 papers. Book: Weste Ch. 2; Kang Ch. 3.

Most often theory (2 of 3). Subtopics: Leakage and DIBL; I-V equations; C-V characteristics.

Formulas:

Subtopic	Formula	Equation	Used in
Leakage and DIBL (5)	**DIBL**	$V_t = V_{t0} - \eta V_{DS}$	19-20 1(d)
	Subthreshold current	$I_{DS} = I_{ds0}\, e^{\frac{V_{GS}-V_t+\eta V_{DS}}{nv_T}} \left(1 - e^{-V_{DS}/v_T}\right)$	19-20 1(d)
	Subthreshold slope	$S = nv_T \ln 10$	19-20 1(d)
I-V equations (8)	Oxide capacitance	$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$	20-21 1(a), 19-20 2(a)
	Gain factor	$\beta = \mu C_{ox} \frac{W}{L}$	19-20 2(a)
	Current, linear	$I_{DS} = \beta \left(V_{GS} - V_t - \frac{V_{DS}}{2}\right) V_{DS}$	19-20 2(a)
	Current, saturation	$I_{DS} = \frac{\beta}{2}(V_{GS} - V_t)^2$	19-20 2(a)
C-V characteristics (4)	**Gate capacitance**	$C_0 = C_{ox} W L$	20-21 1(a)
	Gate capacitance, cutoff	$C_{gb} \approx C_0,\ C_{gs} = C_{gd} = 0$	20-21 1(a)
	Gate capacitance, linear	$C_{gs} = C_{gd} = \frac{1}{2}C_0,\ C_{gb} = 0$	20-21 1(a)
	Gate capacitance, saturation	$C_{gs} = \frac{2}{3}C_0,\ C_{gd} = C_{gb} = 0$	20-21 1(a)

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No.	Paper	Marks	Type	Question

Symbols:

η = DIBL coefficient (about 0.1) ↩

$v_T = kT/q$ = 25.9 mV at 300 K

n = slope factor (1.3–1.5) ↩

C_{ox} = gate capacitance per area (F/m^2) ↩

t_{ox} = oxide thickness (m) ↩

ε_{ox} = permittivity of SiO_2, $3.9\varepsilon_0$ (F/m) ↩

μ = mobility ($m^2/V \cdot s$) ↩

W, L = width, length (m)

C_0 = full gate capacitance (F) ↩

W = gate width (m) ↩

L = gate length (m) ↩

C_{gb} = gate–body capacitance (F) ↩

C_{gs} = gate–source capacitance (F) ↩

C_{gd} = gate–drain capacitance (F) ↩

Questions:

No.	Paper	Marks	Type	Question
Leakage and DIBL				
1	19-20 1(d)	5	theory	Explain the following non-idealities of MOSFET- (i) Drain induced barrier lowering, (ii) subthreshold leakage. What are the effects of temperature on these non-idealities?
I-V equations				
2	19-20 2(a)	8	derivation	Find the equations and I-V characteristics curve of MOSFET current using long-channel model.
C-V characteristics				
3	20-21 1(a)	4	theory	Explain the C-V characteristics of intrinsic MOS gate capacitance in different regions.

4.2. CMOS inverter:

20 marks, 3 questions, in 2 of 2 papers. Book: Weste Ch. 2; Kang Ch. 5, 6.

Most often theory (3 of 3). Subtopics: DC characteristics; Beta ratio; Skew.

Formulas:

Subtopic	Formula	Equation	Used in
DC characteristics (7)	Operating point	$I_{Dn} = \lvert I_{Dp} \rvert$	19-20 3(a)
	Switching threshold	$V_M = \dfrac{V_{tn} + r(V_{DD} - \lvert V_{tp} \rvert)}{1 + r}$, $r = \sqrt{\beta_p/\beta_n}$	19-20 3(a)
	Unity-gain inputs	$V_{IL} = \dfrac{3V_{DD} + 2V_t}{8}$, $V_{IH} = \dfrac{5V_{DD} - 2V_t}{8}$	19-20 3(a)
	Noise margins	$NM_L = V_{IL} - V_{OL}$, $NM_H = V_{OH} - V_{IH}$	20-21 2(b), 19-20 3(a)
Beta ratio (6)	**Switching threshold**	$V_M = \dfrac{V_{DD} - \lvert V_{Tp} \rvert + V_{Tn}\sqrt{\beta_n/\beta_p}}{1 + \sqrt{\beta_n/\beta_p}}$	20-21 2(b)
	Symmetric CMOS V_{IL}	$V_{IL} = \dfrac{3V_{DD} + 2V_T}{8}$, $V_{IH} = \dfrac{5V_{DD} - 2V_T}{8}$	20-21 2(b)
Skew (7)	HI-skew inverter	$g_u = \frac{5}{6}$, $g_d = \frac{5}{3}$	20-21 3(d)
	LO-skew inverter	$g_u = \frac{4}{3}$, $g_d = \frac{2}{3}$	20-21 3(d)

Symbols:

NM_L = low noise margin (V) ↩ V_{IH} = upper unity-gain input (V) ↩ β_n/β_p = beta ratio ↩

NM_H = high noise margin (V) ↩ V_{OH} = output high voltage (V) ↩ g_u = logical effort, rising output ↩

V_{IL} = lower unity-gain input (V) ↩ V_M = input where $V_{out} = V_{in}$ (V) ↩ g_d = logical effort, falling output ↩

Questions:

No.	Paper	Marks	Type	Question
DC characteristics				
1	19-20 3(a)	7	theory	Using I-V characteristics curve of MOSFET, find DC characteristics of CMOS inverter.
Beta ratio				
2	20-21 2(b)	6	theory	Briefly explain what happens to the transfer characteristics and noise margins when beta ratio is increased in a CMOS inverter pair.
Skew				
3	20-21 3(d)	7	theory	Explain the inverter width sizing for favoring the rising edge, falling edge, or achieving an unskewed inverter.

4.3. Fabrication:

30 marks, 6 questions, in 2 of 2 papers. Book: Weste Ch. 3; Plummer Ch. 4, 5, 7, 8, 10.

Most often theory (6 of 6). Subtopics: Etching; Cleaning; Lithography; Diffusion and implantation.

Formulas:

Subtopic	Formula	Equation	Used in
Etching (6)	Plasma dissociation	$CF_4 + e^- \rightarrow CF_3 + F + e^-$	19-20 1(a)
	Silicon etch	$Si + 4F \rightarrow SiF_4 \uparrow$	19-20 1(a)
	Oxide etch (dry)	$SiO_2 + 4F \rightarrow SiF_4 \uparrow + O_2$	19-20 1(a)
	Oxide etch (wet)	$SiO_2 + 6HF \rightarrow H_2SiF_6 + 2H_2O$	19-20 1(a)
	Adding O_2	$O + CF_3 \rightarrow COF_2 + F$	19-20 1(a)
	Adding H_2	$H + F \rightarrow HF$	19-20 1(a)
	Selectivity	$S = \frac{r_{film}}{r_{under}}$	19-20 1(a)
Cleaning (6)	**RCA clean**	SC-1: NH_4OH:H_2O_2:H_2O = 1:1:5; HF 1:50; SC-2: HCl:H_2O_2:H_2O = 1:1:6	19-20 1(b)
Lithography (16)	Lithography resolution	$W_{min} = k_1 \frac{\lambda}{NA}$	19-20 1(c)

No.	Paper	Marks	Type	Question

Symbols:

r = etch rate of each layer (nm/min) NA = numerical aperture ↩

λ = wavelength (m) ↩ k_1 = process factor ↩

Questions:

No.	Paper	Marks	Type	Question
Etching				
1	19-20 1(a)	6	theory	Briefly discuss the chemistries involved in in *[sic: should be "involved in"]* etching. Also explain the effect of adding H_2 and O_2 on etching rate.
Cleaning				
2	19-20 1(b)	6	theory	What is RCA clean? What are the steps? What contaminants can be removed in this process?
Lithography				
3	20-21 5(c)	2	theory	Explain the following: (i) positive mask, (ii) negative mask, (iii) positive photoresist and (iv) negative photoresist.
4	20-21 5(d)	6	theory	Describe the process sequence for patterning SiO_2 as illustrated in the figure 5(d).
5	19-20 1(c)	8	theory	Define photolithography. briefly describe the major steps involved in photolithography.
Diffusion and implantation				
6	20-21 1(e)	2	theory	Differentiate between the diffusion process and the ion implantation process in semiconductor fabrication.

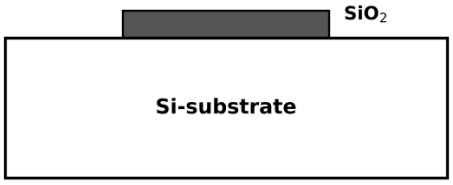


Figure 1: 20-21 5(d): Figure 5(d) of the paper (redrawn)

4.4. Delay and logical effort:

46 marks, 8 questions, in 2 of 2 papers. Book: Weste Ch. 4.

Most often numerical (5 of 8). Subtopics: Gate delay; Path delay; Input ordering; Elmore delay; Sizing.

Formulas:

Subtopic	Formula	Equation	Used in
Gate delay (7)	**Elmore delay**	$t_{pd} = \sum_i R_i C_i$	19-20 2(b)
	NAND3 rising, worst	$t_{pdr} = (9 + 5h)RC$	19-20 2(b)
	NAND3 falling, worst	$t_{pdf} = (12 + 5h)RC$	19-20 2(b)
	NAND3 contamination	$t_{cdr} = \left(3 + \frac{5h}{3}\right) RC,\ t_{cdf} = (9 + 5h)RC$	19-20 2(b)
Path delay (25)	**Gate delay**	$d = gh + p$	20-21 1(d), 19-20 3(b)
	Logical efforts	inv $g = 1$; NAND-n $\frac{n+2}{3}$; NOR-n $\frac{2n+1}{3}$; AOI22 2	19-20 3(b)

Subtopic	Formula	Equation	Used in
	Branching effort	$b = \frac{C_{on} + C_{off}}{C_{on}}$	19-20 3(b)
	Path effort	$F = GBH,\ G = \prod g_i,\ B = \prod b_i,\ H = \frac{C_{out}}{C_{in}}$	20-21 1(d), 19-20 3(b)
	Minimum delay	$\hat{f} = F^{1/N},\ \hat{D} = NF^{1/N} + P$	19-20 3(b)
	Gate size	$C_{in,i} = \frac{g_i C_{out,i}}{\hat{f}}$	20-21 1(d), 19-20 3(b)
	Best stages	$\hat{N} = \log_4 F$	19-20 3(c)
	Logical effort, NAND and NOR	$g_{NAND\text{-}n} = \frac{n+2}{3},\ g_{NOR\text{-}n} = \frac{2n+1}{3}$	20-21 1(d)
	Minimum path delay	$\hat{D} = NF^{1/N} + P,\ \hat{f} = F^{1/N}$	20-21 1(d)
	Best number of stages	$\hat{N} \approx \log_4 F$	20-21 2(c)
	FO4 delay	$d_{FO4} = 1 \cdot 4 + 1 = 5\tau$	20-21 1(d)
Input ordering (5)	Input order (NAND2)	outer $p = 2$, inner $p = 7/3$	19-20 3(d)
Elmore delay (5)	**Elmore delay**	$t_{D,i} = \sum_k R_{ik} C_k$	20-21 2(a)
Sizing (4)	Unit inverter	$W_p = \frac{\mu_n}{\mu_p} W_n$	20-21 3(b)
	Series stack	$W = k\, W_{unit}$	20-21 3(b)
	Path resistance	$\sum_{\text{path}} \frac{R_{unit} W_{unit}}{W_j} \leq R_{unit}$	20-21 3(b)

Symbols:

R_i = resistance shared with the path to node i

h = number of NAND3 loads; electrical effort, C_{out}/C_{in} ↩

g = logical effort ↩

p = parasitic delay (τ) ↩

d = delay of one gate (τ) ↩

C_{off} = load off the path ↩

F = path effort ↩

G = path logical effort ↩

B = path branching effort ↩

H = path electrical effort ↩

b_i = branching of stage i ↩

N = number of stages ↩

$C_{in,i}$ = input capacitance of stage i ↩

$C_{out,i}$ = load of stage i ↩

n = number of inputs ↩

$\hat{D}$ = least path delay (τ) ↩

P = sum of parasitic delays (τ) ↩

$\hat{f}$ = best stage effort ↩

$t_{D,i}$ = delay from the input to node i (s) ↩

C_k = capacitance at node k (F) ↩

R_{ik} = resistance shared by the paths to i and k (Ω) ↩

W_n = nMOS width ↩

W_p = pMOS width for equal rise and fall ↩

k = transistors in series on the path ↩

W_{unit} = width in the unit inverter ↩

R_{unit} = resistance of the unit transistor (Ω) ↩

W_j = width of transistor j on the path ↩

Questions:

No.	Paper	Marks	Type	Question
Gate delay				
1	19-20 2(b)	7	numerical	Find propagation and contamination delay (for both rising and falling) of 3-input NAND gate while it is driving 4 identical NAND gates. Consider that the gate has equal pull-up and pull-down resistance.
Path delay				
2	20-21 1(d)	5	numerical	Estimate the minimum delay of the path from A to B in Figure 1(d) and choose transistor sizes to achieve this delay. The initial NAND2 gate may present a load of 8 λ of transistor width on the input and the output load is equivalent to 45 λ of transistor width.
3	20-21 2(c)	7	design	Design an 8-input AND gate using FO4 in different configurations. Determine which configuration provides the fastest performance.
4	19-20 3(b)	6	numerical	Find minimum delay for path $A \rightarrow B$ in the logic circuit of Figure 3.1. Also find the gate size to achieve this delay.
5	19-20 3(c)	7	design	Figure 3.2 shows a portion of a decoder circuit. The inverter at input drives 8 identical 4-input NAND gate with input capacitance y each. The output of the decoder (NAND gate) is connected to a 32-bit word of register and each register bit presents a load of 3-unit sized transistor. Find best number stages (how many inverters should be added between NAND and load?), stage effort, path delay, NAND gate size.
Input ordering				
6	19-20 3(d)	5	theory	Explain input ordering delay effect.
Elmore delay				
7	20-21 2(a)	5	numerical	For the RC circuit shown in Figure 2(a), compute the Elmore delays from input node to node 7 and from input node to node 5 of this circuit in terms of R and C component values.
Sizing				
8	20-21 3(b)	4	numerical	For the logic function $Y = \overline{(A + BC) \cdot D}$, choose transistor widths to achieve equal rise and fall resistance as a unit inverter. Assume, $\mu_n = 3\mu_p$

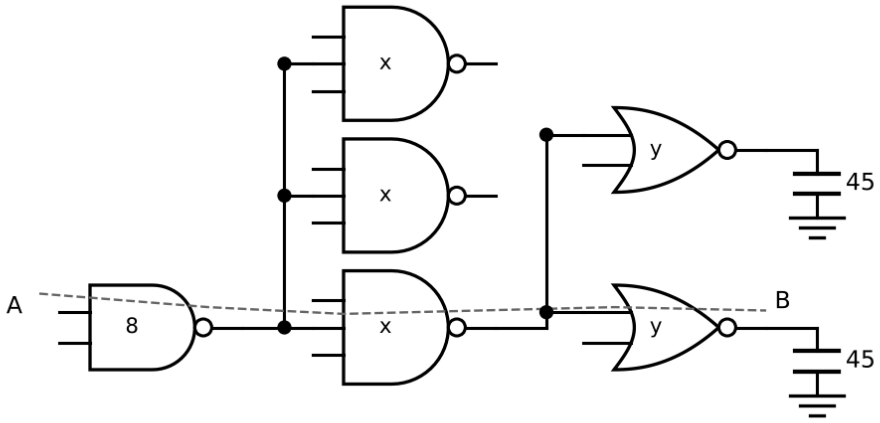


Figure 2: 20-21 1(d): Figure 1(d) of the paper (redrawn)

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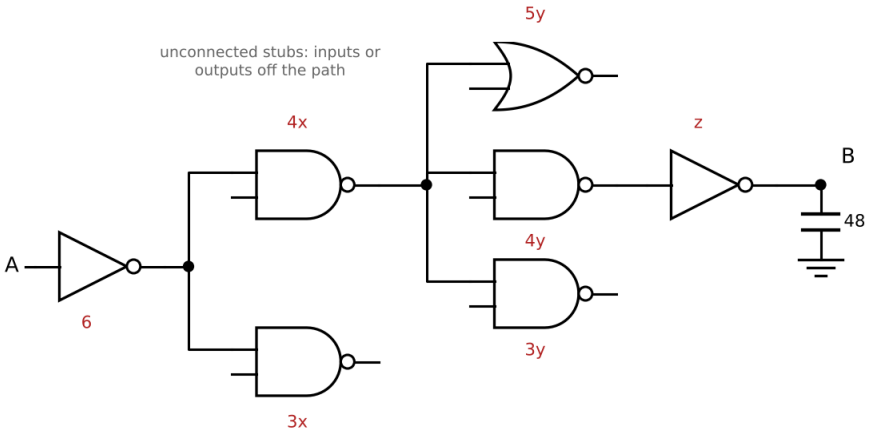


Figure 3: 19-20 3(b): Figure 3.1 of the paper (redrawn)

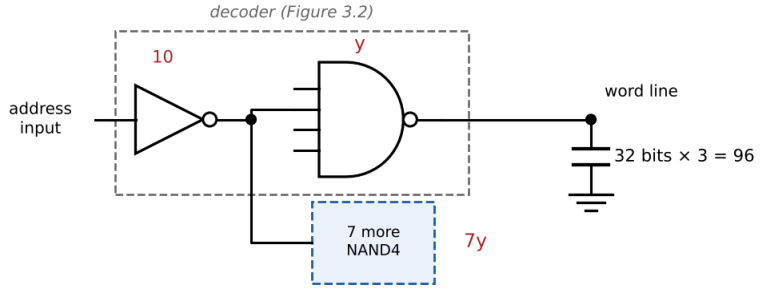


Figure 4: 19-20 3(c): Figure 3.2 of the paper (redrawn)

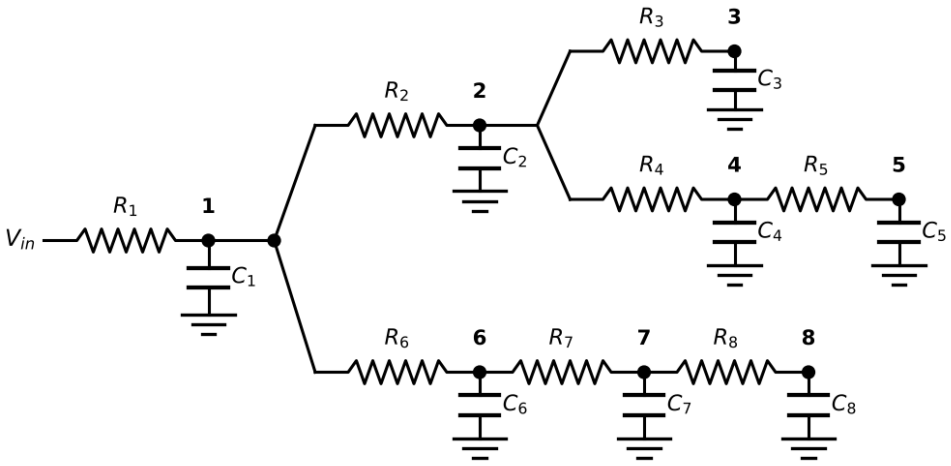


Figure 5: 20-21 2(a): Figure 2(a) of the paper (redrawn)

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4.5. Power:

28 marks, 5 questions, in 2 of 2 papers. Book: Weste Ch. 5.

Most often theory (3 of 5). Subtopics: Glitches; Dynamic power; Leakage.

Formulas:

Subtopic	Formula	Equation	Used in
Dynamic power (16)	**Dynamic power**	$P_{dyn} = \alpha C V_{DD}^2 f$	19-20 4(c)
	Capacitance from area	$C = (C/A) \times A$	19-20 4(c)
	Delay from current	$t_{pd} = \frac{C_L V_{DD}/2}{I_{sat}} = \frac{C_L V_{DD}}{\beta(V_{DD} - V_T)^2}$	20-21 3(c)
	Delay vs supply	$t_{pd} \approx \frac{C_L}{\beta V_{DD}} \propto \frac{1}{V_{DD}}$	20-21 3(c)
	Energy per output cycle	$E = C_L V_{DD}^2$	20-21 3(c)
	Energy stored	$E_C = \frac{1}{2} C_L V_{DD}^2$	20-21 3(c)
	Dynamic power	$P_{dynamic} = \alpha C_L V_{DD}^2 f$	20-21 2(d)
	Static power	$P_{static} = I_{static} V_{DD}$	20-21 2(d)
Leakage (6)	Stack node voltage	$V_x = \frac{\eta V_{DD}}{1 + 2\eta}$	19-20 5(d)
	Stack effect	$\frac{I_{single}}{I_{stack}} = 10^{\frac{\eta V_{DD}(1+\eta)}{(1+2\eta)S}}$	19-20 5(d)

Symbols:

α = activity factor ↩

C = switched capacitance (F) ↩

f = clock frequency (Hz) ↩

t_{pd} = propagation delay (s) ↩

C_L = load capacitance (F) ↩

I_{sat} = saturation current (A) ↩

E = energy taken from the supply (J) ↩

E_C = energy in the charged load (J) ↩

$P_{dynamic}$ = switching power (W) ↩

P_{static} = static power (W) ↩

I_{static} = leakage or contention current (A) ↩

V_x = middle node of two OFF nMOS (V) ↩

Questions:

No.	Paper	Marks	Type	Question
Glitches				
1	19-20 4(b)	6	theory	Define glitch? How does it increase power consumption? How it can be reduced?
Dynamic power				
2	20-21 2(d)	4	theory	Write short note on –i) Static power dissipation ii) Dynamic power dissipation.
3	20-21 3(c)	8	derivation	Show that the dynamic power dissipation of a CMOS inverter varies directly with square of the power supply voltage whereas the delay of the circuit varies inversely with the power supply voltage.
4	19-20 4(c)	4	numerical	You are synthesizing a chip composed of random logic with an average activity factor of 0.1. You are using a standard cell process with an average switching capacitance of 450 pF/mm^2. Estimate the dynamic power consumption of your chip if it has an area of 70 mm^2 and runs at 450 MHz at V_{DD} = 0.9 V.

No.	Paper	Marks	Type	Question
Leakage				
5	19-20 5(d)	6	theory	Define stack effect. Find subthreshold leakage current for two series OFF transistor.

4.6. Interconnect:

10 marks, 2 questions, in 1 of 2 papers. Book: Weste Ch. 6.

Most often theory (2 of 2). Subtopics: Crosstalk; Sheet resistance.

Formulas:

Subtopic	Formula	Equation	Used in
Crosstalk (7)	**Crosstalk, floating victim**	$\Delta V_v = \frac{C_{adj}}{C_{gnd} + C_{adj}} \Delta V_a$	19-20 4(d)
	Crosstalk, driven victim	$\Delta V_v = \frac{C_{adj}}{C_{gnd} + C_{adj}} \cdot \frac{1}{1+k} \Delta V_a,\ k = \frac{\tau_a}{\tau_v}$	19-20 4(d)
	Miller coupling	$C_{eff} = C_{gnd} + k_M C_{adj}$	19-20 4(d)
Sheet resistance (3)	**Sheet resistance**	$R_\square = \frac{\rho}{t},\ R = R_\square \frac{l}{w}$	19-20 4(e)

Symbols:

C_{adj} = coupling capacitance ↩

C_{gnd} = capacitance to ground ↩

τ = RC time constant of each wire

k_M = 0 (same), 1 (quiet), 2 (opposite) ↩

ρ = resistivity ($\Omega \cdot$ m) ↩

t = thickness ↩

l = length ↩

w = width ↩

Questions:

No.	Paper	Marks	Type	Question
Crosstalk				
1	19-20 4(d)	7	theory	Define crosstalk. Show how crosstalk induces noise in different cases. How can crosstalk be reduced?
Sheet resistance				
2	19-20 4(e)	3	theory	Define sheet resistance. How does interconnect affect the performance of a system?

4.7. Scaling:

11 marks, 2 questions, in 1 of 2 papers. Book: Weste Ch. 7.

Most often theory (2 of 2). Subtopics: Moore's law; Scaling rules.

Formulas:

Subtopic	Formula	Equation	Used in
Scaling rules (5)	**Constant-field scaling**	$L, W, t_{ox}, V_{DD}, V_t \times \frac{1}{S};\ N_A \times S$	19-20 4(a)
	Constant-voltage scaling	$L, W, t_{ox} \times \frac{1}{S};\ V$ fixed; $N_A \times S^2$	19-20 4(a)

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Questions:

No.	Paper	Marks	Type	Question
Moore's law				
1	19-20 2(c)	6	theory	State Moore's law. Discuss the problems that arise due to MOSFET scaling.
Scaling rules				
2	19-20 4(a)	5	theory	What is scaling? Explain how different parameters of transistors are scaled?

4.8. Combinational logic styles:

49 marks, 10 questions, in 2 of 2 papers. Book: Weste Ch. 9.

Most often diagram (5 of 10). Subtopics: CVSL; Static gates; Dynamic logic; Stick diagrams; Pseudo-nMOS; Pass transistors.

Formulas:

Subtopic	Formula	Equation	Used in
CVSL (4)	CVSL outputs	$Y = f,\ \overline{Y} = \overline{f}$	19-20 2(d)
Static gates (9)	NAND–NAND form	$AB + CD = \overline{\overline{AB} \cdot \overline{CD}}$	19-20 5(c)
	Compound gate	$\text{AOI22} = \overline{AB + CD}$	19-20 5(c)
	Car alarm	$Y = KB\overline{D} + K\overline{B}D = K(B \oplus D)$	20-21 3(a)
Dynamic logic (9)	Dynamic gate output	$Y = \overline{f}$ in evaluate, $Y = 1$ in precharge	19-20 6(a)
Stick diagrams (12)	**OAI22**	$\overline{f} = \overline{(A + B)(C + D)}$	19-20 6(b)
Pseudo-nMOS (8)	Device gain factor	$k = k'\dfrac{W}{L},\ k' = \mu C_{ox}$	20-21 1(b)
	Drain current, linear	$I_D = \dfrac{k}{2}\left[2(V_{GS} - V_T)V_{DS} - V_{DS}^2\right]$	20-21 1(b)
	Drain current, saturation	$I_D = \dfrac{k}{2}(V_{GS} - V_T)^2$	20-21 1(b)
	Body effect	$V_T = V_{T0} + \gamma\left(\sqrt{\lvert 2\phi_F\rvert + V_{SB}} - \sqrt{\lvert 2\phi_F\rvert}\right)$	20-21 1(b)
	Pseudo-NMOS output low	$k_n\left[2(V_{DD} - V_{Tn})V_{OL} - V_{OL}^2\right] = k_p(V_{DD} - \lvert V_{Tp}\rvert)^2$	20-21 1(b)
	Inverter ratio	$\dfrac{(W/L)_n}{(W/L)_p} = \dfrac{k_n}{k_p} \cdot \dfrac{\mu_p C_{ox}}{\mu_n C_{ox}}$	20-21 1(b)
Pass transistors (7)	2-to-1 multiplexer	$Y = \overline{S}D_0 + SD_1$	20-21 6(a)

Symbols:

K = key in ↩

B = belt fastened ↩

D = door closed ↩

k = device transconductance (A/V^2) ↩

k' = process transconductance (A/V^2) ↩

μ = carrier mobility ($\text{m}^2/\text{V} \cdot \text{s}$) ↩

I_D = drain current (A) ↩

V_{GS} = gate–source voltage (V) ↩

V_{DS} = drain–source voltage (V) ↩

V_T = threshold voltage (V) ↩

V_{T0} = threshold at $V_{SB} = 0$ (V) ↩

γ = body-effect coefficient ($\text{V}^{1/2}$) ↩

ϕ_F = Fermi potential (V) ↩

V_{SB} = source–body voltage (V) ↩

V_{OL} = output low voltage (V) ↩

V_{DD} = supply voltage (V) ↩

k_n = driver gain factor (A/V^2) ↩

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k_p = load gain factor (A/V^2) ↩ $(W/L)_p$ = load aspect ratio ↩ D_0, D_1 = data inputs ↩

$(W/L)_n$ = driver aspect ratio ↩ S = select input

Questions:

No.	Paper	Marks	Type	Question
CVSL				
1	19-20 2(d)	4	theory	Explain the principle of cascode voltage switch logic.
Static gates				
2	20-21 3(a)	0	design	You have to design a car alarm system in which the alarm will ring (Y= 1) if anyone of the following condition is satisfied. (i) The key is in (K= 1), The Seat belt is fastened (B=1) and the door is open (D=0) or (ii) The key is in, Door is closed and Seat belt is not fastened.
3	19-20 5(c)	9	diagram	Design a CMOS logic circuit to compute $F = AB + CD$ using (i) only NAND gates, (ii) compound gate. Choose the transistor width to achieve unit path resistance. If each input can present a maximum of 20 λ of transistor width, output must drive a load equivalent to 100 λ of transistor width, calculate the minimum delay to compute F. Which one is faster? Comment on your answer.
Dynamic logic				
4	20-21 2(e)	3	theory	What is domino gate? Draw the circuit diagram of a domino OR gate.
5	19-20 6(a)	6	theory	Explain the principle of dynamic logic circuit.
Stick diagrams				
6	20-21 1(c)	6	diagram	Draw the schematic and stick diagram of the circuit implementing the logic $Y = \overline{(A + B + C) \cdot D}$
7	19-20 6(b)	6	diagram	Draw the stick diagram of the following Boolean function: $f = (A + B).(C + D)$ with their symbolic diagram and truth table.
Pseudo-nMOS				
8	20-21 1(b)	8	numerical	A pseudo NMOS inverter is designed such that the output voltage becomes 0.25 V when the input voltage is 5 V. The following data are given: $\mu_n C_{ox}$ = 120 μA/V^2, $\mu_p C_{ox}$ = 50 μA/V^2, V_{ton} = 1V, V_{top} = -1 V, V_{DD} = 5 V, γ = 0.5. Assume that the body of the NMOS transistor is connected with the ground and the body of the PMOS transistor is connected with V_{DD}. (i) Draw the circuit diagram of the inverter. (ii) Calculate the inverter ratio i.e., the ratio of driver transistor (W/L) to the load transistor (W/L). (iii) What will be the output voltage if the input voltage is held at 0.25 V? Explain your answer.
Pass transistors				
9	20-21 4(d)	3	diagram	Implement a 2-input AND gate using both transmission gates and pass-transistor logic.
10	20-21 6(a)	4	diagram	Design a 2-to-1 multiplexer using pass transistor logic. Mention its working procedure.

4.9. Sequential circuits and timing:

36 marks, 6 questions, in 2 of 2 papers. Book: Weste Ch. 10.

Most often theory (2 of 6). Subtopics: Hold time; Cycle time; Setup and hold; Flip-flops; FSM.

Formulas:

Subtopic	Formula	Equation	Used in
Hold time (6)	**Min-delay (hold)**	$t_{cd} \geq t_{hold} - t_{ccq}$	19-20 5(a)
	Hold with skew	$t_{cd} \geq t_{hold} - t_{ccq} + t_{skew}$	19-20 5(a)
Cycle time (4)	**Max-delay (setup)**	$T_c \geq t_{pcq} + t_{pd} + t_{setup}$	19-20 5(b)
	Itanium 2 bypass	$T_c = 90 + 1010 + 62 = 1162$ ps	19-20 5(b)
Setup and hold (10)	Latest data arrival	$t_{data,max} = t_{clk1,max} + t_{clk\text{-}q,max} + t_{logic,max}$	20-21 4(a)
	Setup constraint	$t_{data,max} + t_{setup} \leq T + t_{clk2,min}$	20-21 4(a)
	Hold constraint	$t_{data,min} \geq t_{clk2,max} + t_{hold}$	20-21 4(a)
	Maximum frequency	$f_{max} = \frac{1}{T_{min}}$	20-21 4(a)
FSM (6)	Divide-by-3 next state	$S_1' = S_0,\ S_0' = \overline{S_1 + S_0},\ y = \overline{S_1}\,\overline{S_0}$	20-21 6(c)

Symbols:

t_{ccq} = clock-to-Q contamination ↩

t_{cd} = shortest logic delay ↩

t_{pcq} = clock-to-Q propagation ↩

t_{pd} = longest logic delay ↩

$t_{data,max}$ = latest arrival at the capturing D (s) ↩

$t_{clk1,max}$ = latest launch clock (s) ↩

$t_{clk\text{-}q,max}$ = largest clock-to-Q delay (s) ↩

$t_{logic,max}$ = largest logic and wire delay (s) ↩

t_{setup} = setup time (s) ↩

T = clock period (s) ↩

$t_{clk2,min}$ = earliest capture clock (s) ↩

$t_{data,min}$ = earliest arrival at the capturing D (s) ↩

$t_{clk2,max}$ = latest capture clock (s) ↩

t_{hold} = hold time (s) ↩

T_{min} = least period meeting setup (s) ↩

S_1S_0 = state bits

y = counter output ↩

Questions:

No.	Paper	Marks	Type	Question
Hold time				
1	19-20 5(a)	6	theory	What is min-delay constraint? Find min-delay for a system sequenced with flip-flop. If min-delay occurs, what is the solution?
Cycle time				
2	19-20 5(b)	4	numerical	The Itanium 2 ALU self-bypass path is shown in Figure 5.1, and delays associated with each element are given in Table 5.1. Calculate the minimum cycle time T_c at which ALU self-bypass path will operate correctly. The earliest input to the late bypass multiplexer is imm value coming from another flip-flop. Will this path experience any hold-time failures?
Setup and hold				
3	20-21 4(a)	10	numerical	In the following circuit of Figure 4(a), determine whether there is any set up or hold time violation.
Flip-flops				
4	20-21 4(b)	6	diagram	Show the circuit diagram of an asynchronous resettable Flip-Flop and explain its operation utilizing a timing diagram.

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No.	Paper	Marks	Type	Question
5	20-21 6(f)	4	theory	Fill in the truth table for the flip flop shown in Figure 6(f). Is this a positive edge triggered flip flop?
FSM				
6	20-21 6(c)	6	design	You have to design a divide-by-3 counter using FSM. (i) Draw the state transition diagram of the finite state machine described. (ii) Derive the state transition table and equations for the next-state logic. (iii) Draw the complete FSM circuit.

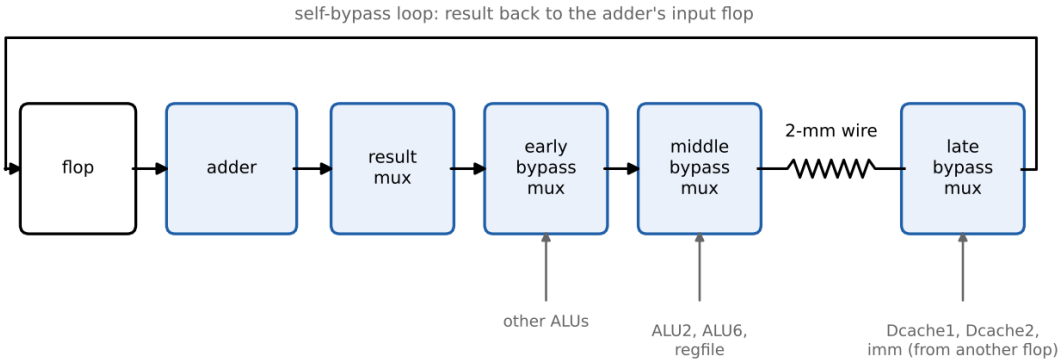


Figure 6: 19-20 5(b): Figure 5.1 of the paper (redrawn)

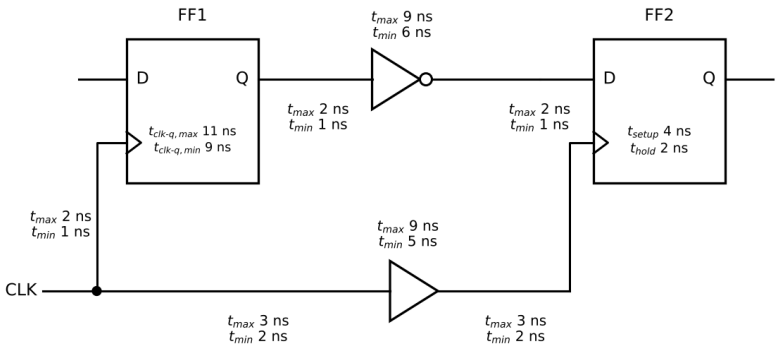


Figure 7: 20-21 4(a): Figure 4(a) of the paper (redrawn)

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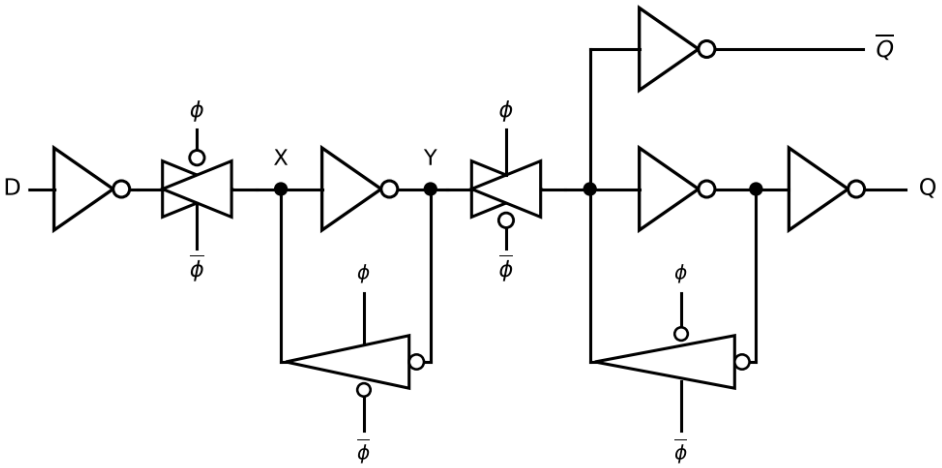


Figure 8: 20-21 6(f): Figure 6(f) of the paper (redrawn)

ϕ	$\overline{\phi}$	D	X	Y	$\overline{Q}$	Q
0	1		1			1
1	0	1				
0	1	1				
1	0	1				

Figure 9: 20-21 6(f): The table of Figure 6(f), as printed

4.10. Datapath:

7 marks, 1 question, in 1 of 2 papers. Book: Weste Ch. 11.

Most often theory (1 of 1). Subtopics: Bus architectures.

Questions:

No.	Paper	Marks	Type	Question
Bus architectures				
1	20-21 5(b)	7	theory	A 4-bit datapath consists of register, ALU and a shifter. Show two possible bus architectures of the system such that an addition operation of two operands stored in the register and storing the result back in the register can be computed in at most two clock cycles.

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4.11. Memories and arrays:

26 marks, 4 questions, in 2 of 2 papers. Book: Weste Ch. 12.

Most often diagram (3 of 4). Subtopics: SRAM; ROM; PLA.

Formulas:

Subtopic	Formula	Equation	Used in
SRAM (16)	SRAM sizing	pull-down > access > pull-up	19-20 6(c)
	Bit-line read time	$t = \frac{C_{bit}\,\Delta V}{I_{cell}}$	20-21 5(a)
ROM (7)	**NOR ROM**	bit line $= \overline{\sum \text{word lines with a transistor}}$	19-20 6(d)
PLA (3)	XOR as sum of products	$A \oplus B = A\overline{B} + \overline{A}B$	20-21 6(e)

Symbols:

C_{bit} = bit-line capacitance (F) ↩ I_{cell} = cell read current (A) ↩

ΔV = swing needed (V) ↩ A, B = inputs ↩

Questions:

No.	Paper	Marks	Type	Question
SRAM				
1	20-21 5(a)	10	diagram	(i) Show the circuit diagram of a 2 x 2 SRAM array which uses 6-transistor memory cell as basic storage cell. Clearly show the row select, column select, pre-charge and sense signal in your circuit. (ii) Explain how the READ and WRITE operation are performed showing the second row and first column cell as an example. (iii) Explain how sense amplifier reduces the READ time of the cell.
2	19-20 6(c)	6	theory	Draw logic diagram of a SRAM cell and explain its different modes of operation.
ROM				
3	19-20 6(d)	7	diagram	Design and implement a pseudo-nMOS ROM to store the data shown in Figure 6.1. Also draw the dot diagram representation.
PLA				
4	20-21 6(e)	3	diagram	Sketch a dot diagram for a 2-input XOR using a PLA.

4.12. Verilog and design flow:

14 marks, 3 questions, in 1 of 2 papers. Book: Weste Appendix A.

Most often design (2 of 3). Subtopics: Design flow; Verilog.

Questions:

No.	Paper	Marks	Type	Question
Design flow				
1	20-21 4(c)	6	theory	Suppose you have written Verilog code to operate a microprocessor. What will be the upcoming steps till its fabrication? Briefly explain your answer.
Verilog				
2	20-21 6(b)	5	design	Design a 7-segment display driver in Verilog.
3	20-21 6(d)	3	design	Write a Verilog code for the FSM described in Question 6(c).

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5. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
MOS transistor theory				
1	19-20 2(a)	8	I-V equations	Write the channel charge and electron speed; integrate along the channel; set the pinch-off condition.
Power				
2	20-21 3(c)	8	Dynamic power	Add the energy per cycle for dynamic power; divide the capacitor charge by the saturation current for delay.

6. Repeat tracker:

No repeats among these papers.

7. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical	Diagram	Design
MOS transistor theory	2	1	·	·	·
CMOS inverter	3	·	·	·	·
Fabrication	6	·	·	·	·
Delay and logical effort	1	·	5	·	2
Power	3	1	1	·	·
Interconnect	2	·	·	·	·
Scaling	2	·	·	·	·
Combinational logic styles	3	·	1	5	1
Sequential circuits and timing	2	·	2	1	1
Datapath	1	·	·	·	·
Memories and arrays	1	·	·	3	·
Verilog and design flow	1	·	·	·	2

8. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Combinational logic styles	49	17 %
2	Delay and logical effort	46	32 %
3	Sequential circuits and timing	36	45 %
4	Fabrication	30	55 %
5	Power	28	64 %
6	Memories and arrays	26	73 %
7	CMOS inverter	20	80 %
8	MOS transistor theory	17	86 %
9	Verilog and design flow	14	90 %
10	Scaling	11	94 %
11	Interconnect	10	98 %
12	Datapath	7	100 %

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